

1.1 - Limits

Goals: Describe limits graphically, numerically, and algebraically.

Before Class Videos: 1.1.1 and 1.1.2

Example 1: Consider these three functions:

$$f(x) = x + 1, \quad g(x) = \frac{x^2 - 1}{x - 1} = \frac{(x + 1)(x - 1)}{x - 1}, \quad h(x) = \begin{cases} x + 1, & x \neq 1, \\ 3, & x = 1. \end{cases}$$

How are these similar or different?

Definition:

$\lim_{x \rightarrow a} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently close to a .

Example 2: Compute the limit as $x \rightarrow 1$ of $f(x)$, $g(x)$, and $h(x)$.

Example 3: Use a table of values to estimate $\lim_{x \rightarrow 1} (x + 1)$.

Example 4: Use a table of values to estimate $\lim_{x \rightarrow 0} \frac{1}{x}$.

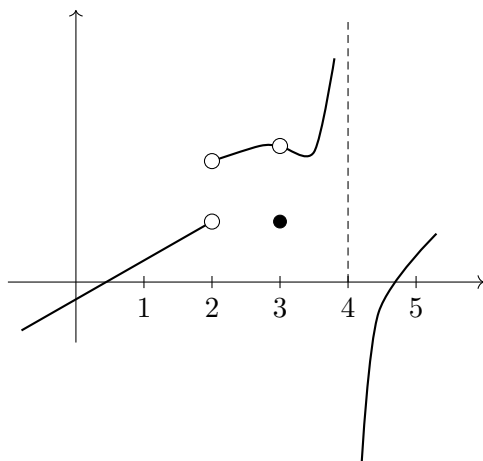
Example 5: Investigate

$$f(x) = \begin{cases} x, & x < 1, \\ x + 1, & x \geq 1. \end{cases}$$

Definition:

$\lim_{x \rightarrow a^+} f(x)$ means $f(x)$ gets arbitrarily close to L for all x sufficiently close to a from the right,
i.e. $x > a$.

Example 6: Use the graph to evaluate the indicated limits.



$$\lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 2} f(x) = \underline{\hspace{2cm}},$$

$$\lim_{x \rightarrow 3^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 3^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 3} f(x) = \underline{\hspace{2cm}},$$

$$\lim_{x \rightarrow 4^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 4^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 4} f(x) = \underline{\hspace{2cm}}.$$

Example 7: Investigate limits of

$$f(x) = \sin\left(\frac{1}{x}\right)$$

Example 8: Graphically investigate limits of $\frac{x^2-1}{x^2-x-2}$